SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence   COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%   QUESTIONS: 3   Duration: 1 Hour   Total Marks: 30

ANNEXURE A

Constraint-Based Problem Solving

Code of Conduct:

I Bharath P (Reg No: 192511282) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Bharath

QUESTIONS

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

i. D1 cannot work Night shift

ii. D2 must work before D3 (Morning < Afternoon < Night)

iii. Only one doctor per shift

iv. D3 cannot work Morning

v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

i. Movement allowed: Up, Down, Left, Right

ii. Cost of each move = 1

iii. Cannot pass through obstacles

iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)

• Each movement has a cost of 1, but entering risky zones has an additional cost of 2

• The robot must minimize total cost while ensuring safe navigation

• The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.

c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

ANSWERS

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts

1. Hospital Doctor Shift Assignment — Backtracking Search

Given constraints

- D1 cannot work Night.
- D2 must work before D3.
- Only one doctor per shift.
- D3 cannot work Morning.
- Every doctor must receive exactly one shift.

Backtracking steps

Step 1: Assign D1

Try:

D1 → Morning

Constraint check:

- D1 is not assigned Night → **Valid**

Current assignment:

Shift	Doctor
Morning	D1
Afternoon	—
Night	—

Step 2: Assign D2

Try:

D2 → Afternoon

Constraint check:

- D2 must work before D3.
- Therefore, D3 can be assigned Night.
- Only one doctor occupies Afternoon → **Valid**

Current assignment:

Shift	**Doctor**
Morning	D1
Afternoon	D2
Night	—

Step 3: Assign D3

Assign:

D3 → Night

Constraint check:

- D3 is not assigned Morning → **Valid**
- D2 = Afternoon and D3 = Night → D2 works before D3 → **Valid**
- Only one doctor per shift → **Valid**
- Every doctor has exactly one shift → **Valid**

Final valid schedule

Shift	**Doctor**
Morning	**D1**
Afternoon	**D2**
Night	**D3**

Final Answer:

D1 → Morning, D2 → Afternoon, D3 → Night

2. Robot Grid

The uploaded question does not provide the actual 5×5 grid, the coordinates of S and G, or the locations of the obstacles. Therefore, the exact node-expansion sequence, priority-queue contents, optimal path, and numerical cost cannot be determined from the supplied question.

BFS solution procedure:

1. Put Start node S into the queue.

2. Expand S and generate valid Up, Down, Left and Right neighbours.

3. Reject cells outside the 5×5 grid, blocked cells, and already visited cells.

4. Insert each newly discovered valid cell into the FIFO queue.

5. Expand nodes in FIFO order until Goal G is reached.

6. Store the parent of each discovered node.

7. Trace the parent links from G back to S to obtain the shortest path.

8. Since every movement has cost 1, Path Cost = number of moves.

Manhattan Distance:

For a cell (x, y) and goal (xg, yg), Manhattan Distance = $|x - xg| + |y - yg|$.

Note: Manhattan Distance is a heuristic. Standard BFS itself expands nodes according to FIFO order and does not use the heuristic to prioritize its queue. An exact numerical path requires the missing grid and obstacle layout.

The question specifies a **5×5 grid**, four-directional movement, movement cost of 1, obstacles, and Manhattan Distance heuristic. However, the uploaded question **does not provide the actual positions of S, G, or the obstacles**.

Therefore, an exact numerical path and node-expansion sequence cannot be determined from the given information.

BFS procedure

1. Start from node **S**.
2. Put S into the queue.
3. Expand S and examine:
    - Up
    - Down
    - Left
    - Right
4. Remove cells outside the grid.
5. Remove cells containing obstacles.
6. Remove already visited cells.
7. Add valid neighbouring cells to the queue.
8. Continue expanding nodes in **FIFO order**.
9. Stop when **G** is reached.
10. Trace the parent nodes from G back to S to obtain the path.

Manhattan Distance

For a current cell (x, y) and goal (xg, yg):

Manhattan Distance = $|x - xg| + |y - yg|$

Since every movement costs 1, BFS finds the shortest path in terms of the **number of moves**, provided the grid is unweighted.

Cost

Path Cost = Number of movements × 1

So, without the missing grid layout, the exact optimal path and numerical cost **cannot be calculated honestly**.

3. Autonomous Rescue Robot

The robot operates in a partially observable environment, can move in four directions, must avoid obstacles, has movement and risk costs, and updates its knowledge dynamically.

a) Analysis of the constraints

1. Movement constraint

The robot can move only Up, Down, Left, and Right. Diagonal movement is not allowed.

2. Obstacle constraint

Blocked cells cannot be entered. The robot must identify and avoid these cells.

3. Partial observability

The robot can observe only adjacent cells. Therefore, it does not initially know the complete environment.

4. Movement cost

Every normal movement has a cost of:

Cost = 1

5. Risk-zone cost

Entering a risky zone adds an additional cost of 2.

Therefore:

Total cost for risky movement = 1 + 2 = 3

6. Minimum-cost requirement

The robot must reach the survivor while minimizing the total movement cost.

7. Dynamic knowledge

The robot must continuously update its knowledge as it discovers new obstacles, safe areas, and risky zones.

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b) Solution approach — Uniform Cost Search / Online Search

Step 1: Start at position S with cost 0.

Step 2: Sense the adjacent cells.

Step 3: Update the robot's internal map.

Step 4: Generate all reachable neighbouring cells.

Step 5: Calculate the cost of moving to each cell.

- **Normal cell → cost = 1**
- **Risky cell → cost = 3**

Step 6: Add the available cells to the frontier.

Step 7: Uniform Cost Search selects the node having the lowest accumulated path cost.

Step 8: Continue exploring while avoiding obstacles.

Step 9: Whenever new information is detected, update the internal map and reconsider the available paths.

Step 10: Continue until the robot reaches G.

Thus, the robot obtains a minimum-cost route based on the information it has discovered.

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c) Application of AI concepts

Intelligent Agent

The rescue robot is an intelligent agent because it:

- **Perceives the environment using sensors.**
- **Processes the available information.**
- **Selects actions.**

- **Moves toward the survivor.**
- **Updates its knowledge based on new observations.**

Rationality

The robot behaves rationally by selecting actions that help it:

- **Reach the survivor.**
- **Avoid obstacles.**
- **Avoid unnecessary risky zones.**
- **Minimize the total path cost.**
- **Maintain safe navigation.**

Environment Type

The environment can be classified as:

- **Partially observable — the robot sees only adjacent cells.**
- **Discrete — the environment consists of individual grid cells.**
- **Sequential — each action affects future decisions.**
- **Dynamic from the agent's knowledge perspective — the robot's map changes as it discovers new information.**

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d) Justification of the approach

Uniform Cost Search (UCS) is suitable because different movements can have different total costs.

A normal movement costs 1, while entering a risky zone costs:

1 + 2 = 3

UCS expands the node with the smallest accumulated cost first. Therefore, it can select a cheaper route even when that route contains more steps than another expensive route.

Since the robot has limited sensing, an Online Search Agent is also appropriate. The robot can explore the environment, update its knowledge, and change its route when new obstacles or risky zones are discovered.